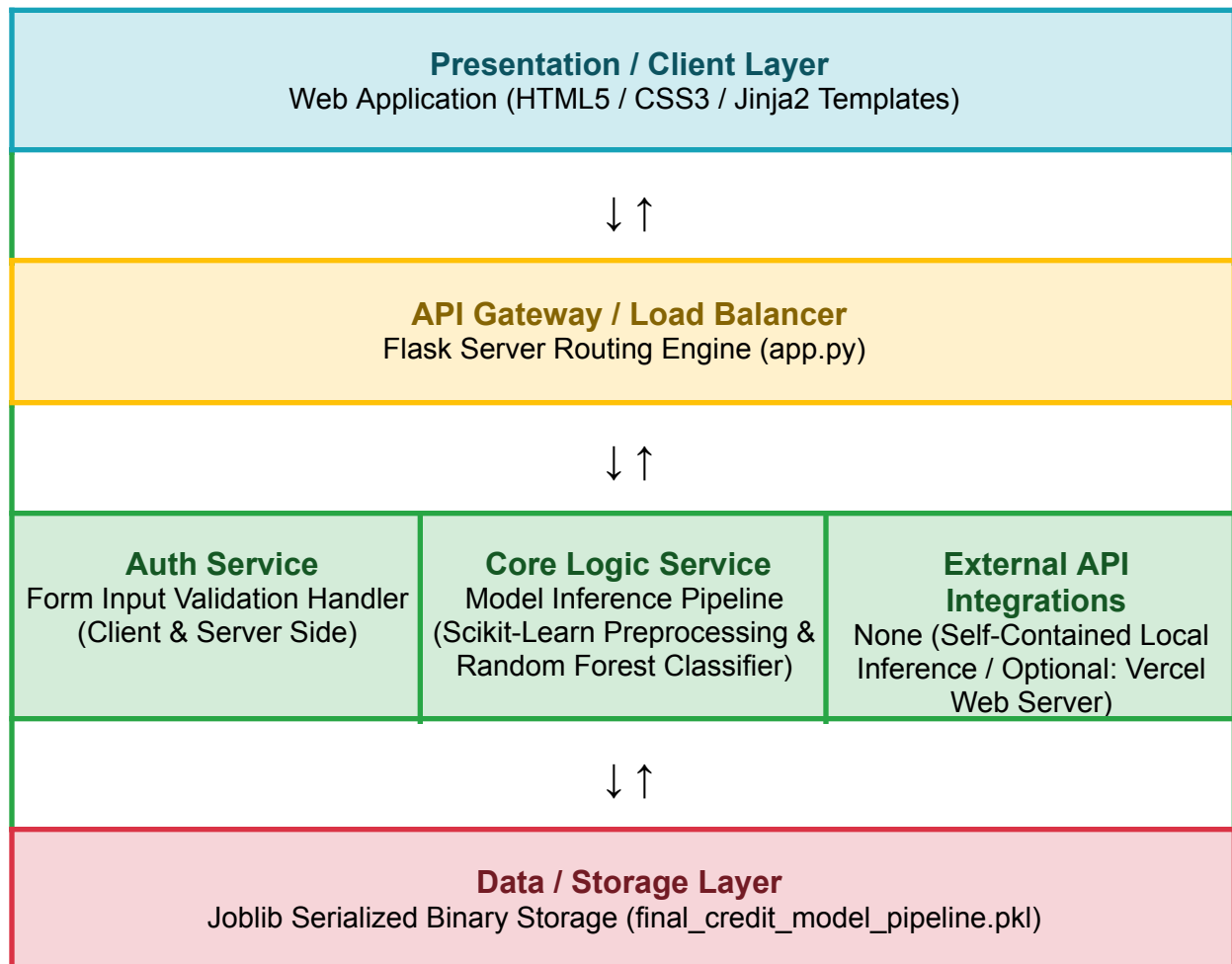


Solution Architecture

Date	15 March 2026
Team ID	XXXXXX
Project Name	Credit Card Auto-Approval Prediction System
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders the interactive frontend interface. Displays the credit card eligibility form (inputs for income, age, employment status, assets) and visualizes predictions (Approved/Rejected) and confidence ratings.	HTML5, CSS3, Jinja2 template rendering.
[API Gateway	Directs HTTP traffic, intercepts form submission POST requests, routes inputs to predictive handlers, and sends processed output variables back to HTML views.	Flask server routing (app.py).
Microservices / Core Logic	Handles data normalization, label encoding for categoricals, numerical scaling, and processes clean data arrays through the Random Forest model to calculate decisions.	Python, Pandas, Numpy, Scikit-Learn.
[Database / Storage	Stores the fully trained, binary-serialized machine learning pipeline including preprocessing rules and classifier weights.	Joblib Serialization (.pkl file format).